AHHYUN YUH

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EDUCATION

Sep 2022 - Present	**University of Pennsylvania** *PhD Candidate in Computer and Information Science*	Philadelphia, PA, USA
Mar 2020 - Feb 2022	**Kyungpook National University** *Master of Science in Electronic Engineering*	Daegu, Korea
Mar 2015 - Feb 2020	**Kyungpook National University** *Bachelor of Science in Electronic Engineering*	Daegu, Korea
Feb 2018 - Aug 2018	**University of Ljubljana** *Exchange Student Program*	Ljubljana, Slovenia

PROJECTS

iCareChat: Voice-Based LLM Agent for Older Adults (Ongoing, Lead Developer)
Designed and currently leading the development of iCareChat, a privacy-preserving, HIPAA-compliant, voice-based chatbot application using large language models. Built with a multi-agent framework and real-time audio processing pipeline to deliver empathetic, accessible support for older adults. Evaluated via IRB-approved clinical studies across the US and Japan to assess mental health outcomes and usability.

TrueDepth 3D Face Reconstruction for Graves' Disease Monitoring (Ongoing)
Developing an iOS-based facial monitoring tool using Apple's ARKit and TrueDepth camera to track morphological changes in patients with Graves' disease. Combines RGB-D alignment, landmark detection, and privacy-focused data handling to enable in-home monitoring integrated into clinical workflows.

iCareLoop: Sensing Framework for Detecting Social Isolation and Loneliness
Led the end-to-end design and implementation of iCareLoop, an IoT-based, passive monitoring system to identify social isolation and loneliness in older adults through behavioral signal analysis. Applied time-series modeling and pattern recognition to sensor data from US and Japanese eldercare settings. System validated in multi-country deployment studies and published across major health-tech venues.

BlinkWise: mmWave-Based Mental State Monitoring via Blink Dynamics
Co-developed BlinkWise, a low-power wearable device using mmWave radar and ESP32 microcontroller to monitor blink patterns and infer real-time cognitive states. Designed custom signal processing pipeline (FFT, beamforming, adaptive filtering) for high-accuracy, low-latency blink detection. System supports applications in mental workload tracking and user-context inference.

Real-Time Sound Event Classification for Human Activity Recognition
Developed an edge-based ambient sound classification system for real-time, privacy-conscious human activity recognition in smart homes. Designed lightweight audio feature extractors and trained a CNN-based classifier deployable on resource-constrained embedded devices. Achieved robust performance for daily living activity monitoring with no camera-based intrusion.

PUBLICATIONS

1. "IoT-Based Loneliness Monitoring for Older Adults: Lessons from Japanese and US Communities", IEEE Access, submitted, 2025
2. "SMILE: Sensor Data-driven Detection and Proximate Cause Analysis for Loneliness in Older Adults", *ACM Conference on Health, Inference, and Learning (ACM Health)*, submitted, 2025

3. "Tracking Blink Dynamics and Mental States on Glasses", *Proceedings of the ACM on Mobile Systems, Applications, and Services (MobiSys)*, accepted, 2025
4. "Older Adults' Perceptions and Attitudes Toward Passive Sensors to Measure Loneliness: A Qualitative Study", *SAGE Open Aging*, 2025
5. "Perceptions and Attitudes Toward Loneliness Detection using Passive Sensing Technologies in Older Adults", *Innovation in Aging*, Vol. 8, Suppl. 1, 2024
6. "Real-Time Prediction of Resident ADL Using Edge-Based Time-Series Ambient Sound Recognition", *Sensors*, Vol. 24, No. 19, 2024
7. "Exploring Effective Sensing Indicators of Loneliness for Elderly Community in US and Japan", *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*, 2024
8. "iCareLoop: Data Management System for Monitoring Gerontological Social Isolation and Loneliness", *Proceedings of the 14th International Conference on Mobile Computing and Ubiquitous Networking (ICMU)*, 2023
9. "Short: Integrated sensing platform for detecting social isolation and loneliness in the elderly community", *IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, 2023
10. "iCareLoop: Closed-Loop Sensing and Intervention for Gerontological Social Isolation and Loneliness", *Proceedings of the ACM/IEEE 14th International Conference on Cyber-Physical Systems (ICCPS)*, 2023
11. "Real-Time Sound Event Classification for Human Activity of Daily Living using Deep Neural Network", *Proceedings of the 14th IEEE International Conference on Internet of Things (iThings)*, Melbourne, 2021
12. "Self-Organized Sensor Fusion between IoT Devices for Resident Activity Recognition and Evaluation System", *Proceedings of the Institute of Embedded Engineering of Korea Conference*, in Korean, 2020

WORK EXPERIENCE

May 2020 - Feb 2022

MATLAB Student Ambassador — MathWorks Korea

https://www.facebook.com/groups/KNUMATLAB

- Led technical workshops and hands-on MATLAB/Simulink sessions, driving adoption and proficiency among engineering students.
- Developed social media strategy and content for the Kyungpook National University MATLAB community, increasing engagement across Facebook and Instagram.
- Acted as a technical mentor and liaison between students and MathWorks, promoting MATLAB in curriculum and extracurricular projects.
- Strengthened communication and presentation skills by teaching complex software tools to diverse student groups.

Mar 2020 - Feb 2022

Graduate Research Assistant – Center for Self-Organizing Software — Kyungpook National University

http://www.csosp.org/

- Contributed to the Self-Organized Ambient Assisted Living (SAAL) initiative, designing intelligent IoT systems to enhance in-home care.
- Implemented sensor fusion frameworks and adaptive system control techniques for real-world deployment in smart environments.
- Collaborated with faculty and engineers to translate academic prototypes into functional, testable solutions for eldercare.
- Gained experience in system architecture, data handling, and cross-disciplinary research in embedded software applications.

Sep 2019 - Feb 2022

Undergraduate Researcher – Real-Time Systems Lab — Kyungpook National University

https://sites.google.com/view/knu-rtlab/

- Designed and implemented real-time audio classification systems for activity recognition in smart home environments.
- Conducted performance evaluations and co-authored academic papers, contributing to award-winning research recognized at KNU-EE Congress.
- Built expertise in embedded systems, machine learning model deployment, and low-latency signal processing.

- Demonstrated strong research-to-product thinking by developing end-to-end solutions suitable for aging-in-place technologies.

Mar 2018 - Feb 2019 **Technical Intern – RLS Merilna Tehnika (Industrial Sensors)** Vodice, Slovenia

https://www.rls.si/

- Supported international product marketing and localization strategy for the Korean market.
- Translated technical documentation (e.g., datasheets, product manuals, web content) from English to Korean with high technical accuracy.
- Assisted in preparing for SPS Germany 2018, contributing to RLS's international outreach and engineering demonstrations.
- Developed cross-cultural communication and global teamwork skills by working across engineering and business teams in Europe and Asia.

AWARDS AND HONORS

2nd Place, KNU-EE Research Congress – Kyungpook National University, 2021
Recognized for research on real-time sound event classification for activity recognition using deep neural networks.

2nd Place, Startup Idea & Pitching Competition – Gyeongsangbuk-do Creative Content Agency, 2019
Awarded for proposing an innovative e-commerce business model integrating virtual fabric printing and supply chain optimization.

1st Place, Intelligent Robot Tournament – Kyungpook National University, 2015
Designed and programmed a pick-and-place robot to complete dynamic tasks in a competitive environment.